

1.0 Basic Algebra

MA1905

1.1 Define the terms:

1.1.1 Constants

1.1.2 Variables

1.1.3 Equations

1.1.4 Functions

Constant: A quantity whose value does not change

For example, in the expression $4x + 12$, the number 12 is a constant.

Variable: A letter or symbol representing a value that can change. Common variables include x and y .

In the expression $4x + 12$, x is the variable.

For example, if $x = 5$, then $4(5) + 12 = 20 + 12 = 32$.

Equation: A mathematical statement showing that two expressions are equal. An equation contains an equal sign ($=$).

For example, $3x + 5 = 20$ is an equation. However, $3x + 5$ is an expression, not an equation.

Function: A relationship where each input has exactly one output. A function is often written using $f(x)$.

For example, if $f(x) = 2x + 3$, then $f(4) = ?$

Step 1: Replace x with 4.

$$f(4) = 2(4) + 3$$

Step 2: Calculate.

$$f(4) = 8 + 3 = 11$$

Therefore, $f(4) = 11$.

Think of a function like a simple process in a computer.

Input $\rightarrow$ Rule/Process $\rightarrow$ Output

Examples

Identify the variable and constants in $7x + 15 = 50$.

If $y = 8$, calculate $3y + 4$

For $f(x) = 3x - 2$, find $f(5)$.

Is $6x + 4$ an equation?

1.2 Express values using exponents and scientific notations

Exponents

An exponent tells us how many times a number or variable is multiplied by itself.

$$2^4 = 2 \times 2 \times 2 \times 2 = 16$$

In words, we would say “2 to the power (exponent) of 4”.

The 2 is the base.

The 4 is the exponent.

Write in exponent form.

$$2 \times 2 \times 2$$

$$4 \times 4 \times 4 \times 4 \times 4$$

$$(-5) \times (-5)$$

$$(-x)(-x)(-x)$$

$$3 \cdot x \cdot y \cdot y \cdot 3 \cdot x \cdot x \cdot x$$

$$a \cdot b \cdot b \cdot a \cdot b$$

Zero Power Rule

Any number or expression raised to the zero power will always be 1.

$$x^0 = 1$$

$$10^0 = 1$$

Why is this? Let's list some powers of 3, starting at 3^4 , and then taking one away from the exponent each time. We will stop at an exponent of 0.

Negative Exponent Rule

Let's review and continue our list from earlier starting at 3^4 , and then taking one away from the exponent each time. This time, we will not stop at 0.

Generally, if we think of our expression as a fraction, negative exponents in the numerator must be moved to the denominator. Similarly, negative exponents in the denominator need to be moved to the numerator. When the base with the exponent moves, the exponent is now positive.

$$x^{-1} = \frac{1}{x}$$

$$10^{-1} = \frac{1}{10} = 0.1$$

Laws of Exponents

The Product Rule

When we multiply exponential expressions with the same base, we simplify by adding the exponents.

$$x^a \cdot x^b = x^{a+b}$$

Evaluate.

$$3^4 \times 3^7$$

$$5^8 \cdot 5^{-6}$$

Simplify.

$$x^2 \cdot x^4$$

The Quotient Rule

When we divide exponential expressions with the same base, we simplify by subtracting the exponents ($x \neq 0$ since division by zero is undefined).

If the larger exponent is in the numerator: $\frac{x^a}{x^b} = x^{a-b}$

If the larger exponent is in the denominator: $\frac{x^a}{x^b} = \frac{1}{x^{b-a}}$

Evaluate.

$$\frac{2^7}{2^5}$$

$$\frac{3^4}{3^5}$$

$$\frac{x^6}{x^6}$$

The Power Rule

To raise a power to a power, keep the same base and multiply the exponents.

$$(x^a)^b = x^{ab}$$

Evaluate.

$$(2^2)^3$$

$$(x^{-3})^5$$

Scientific Notation

Scientific notation is a convenient way to write very large or very small numbers in decimal form using powers of 10.

To write numbers in scientific notation:

- Move the decimal point so there is one nonzero digit to the left of the decimal point.
- Multiply the result by a power of ten with an exponent equal to the number of places the decimal point was moved. The exponent is positive if the decimal point has been moved to the left. The exponent is negative if the decimal point has been moved to the right.

A number is written in scientific notation if it is in the form $A \times 10^n$, where $1 \leq A < 10$ and n is an integer.

For example,

$$310\,000\,000 = 3.1 \times 10^8$$

$$0.0000042 = 4.2 \times 10^{-6}$$

Convert the numbers from standard form to scientific notation.

4 567 000

0.000461

157 000 000

Convert the numbers from scientific notation to standard form.

2.45×10^{-2}

3.1426×10^3